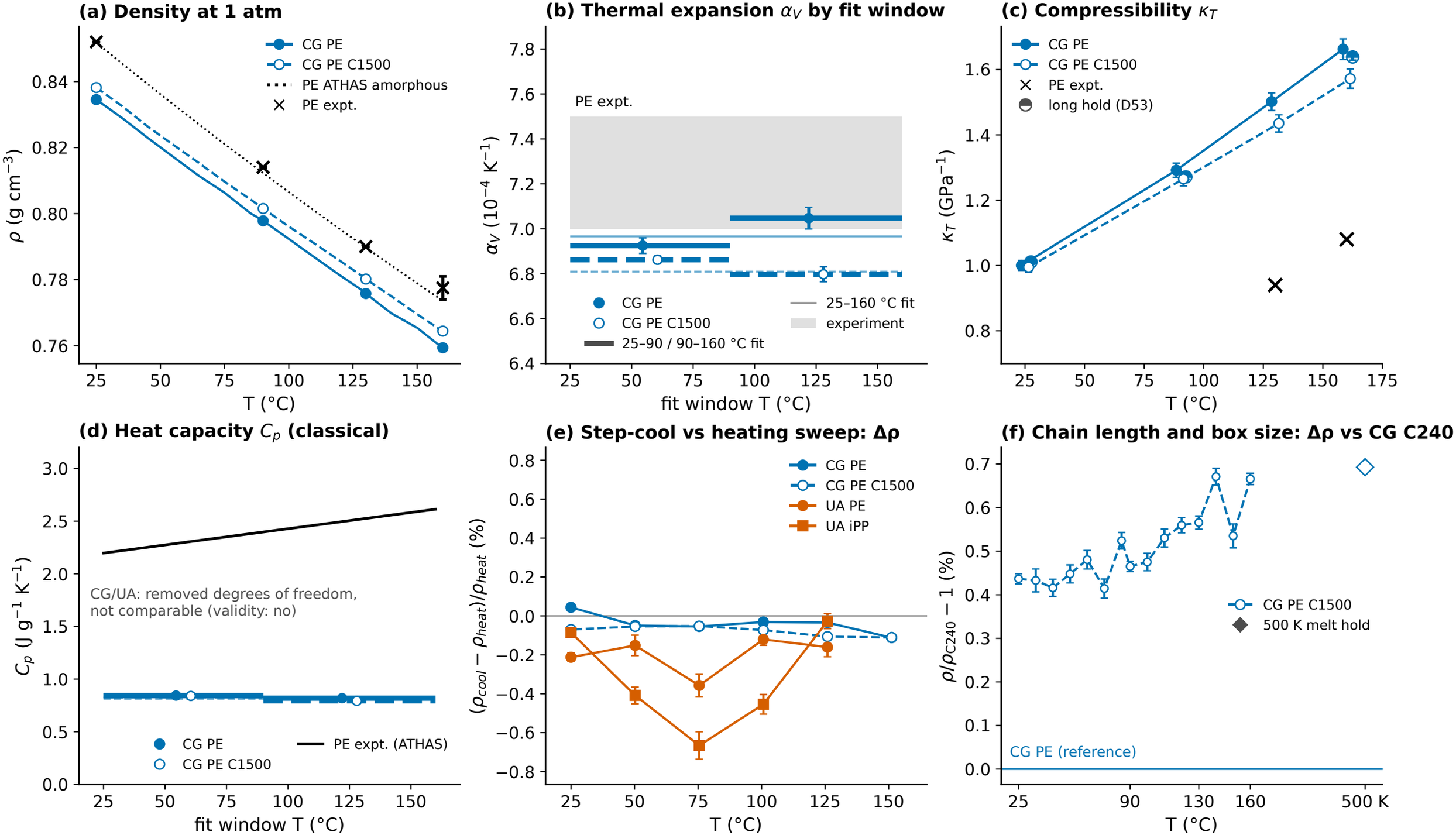

(a) Density at 1 atm
CG PE
CG PE C1500
PE ATHAS amorphous
PE expt.
ρ (g cm⁻³)
T (°C)
(b) Thermal expansion α_V by fit window
PE expt.
CG PE
CG PE C1500
25–90 / 90–160 °C fit
25–160 °C fit
experiment
α_V (10⁻⁴ K⁻¹)
fit window T (°C)
(c) Compressibility κ_T
CG PE
CG PE C1500
PE expt.
long hold (D53)
κ_T (GPa⁻¹)
T (°C)
(d) Heat capacity C_p (classical)
CG/UA: removed degrees of freedom,
not comparable (validity: no)
CG PE
CG PE C1500
PE expt. (ATHAS)
C_p (J g⁻¹ K⁻¹)
fit window T (°C)
(e) Step-cool vs heating sweep: Δρ
CG PE
CG PE C1500
UA PE
UA iPP
(ρ_cool − ρ_heat)/ρ_heat (%)
T (°C)
(f) Chain length and box size: Δρ vs CG C240
CG PE C1500
500 K melt hold
CG PE (reference)
ρ/ρ_C240 − 1 (%)
T (°C)